VLM

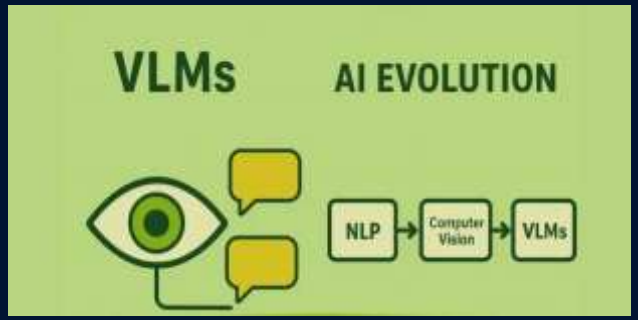
Sayed Mostafa Modaghegh

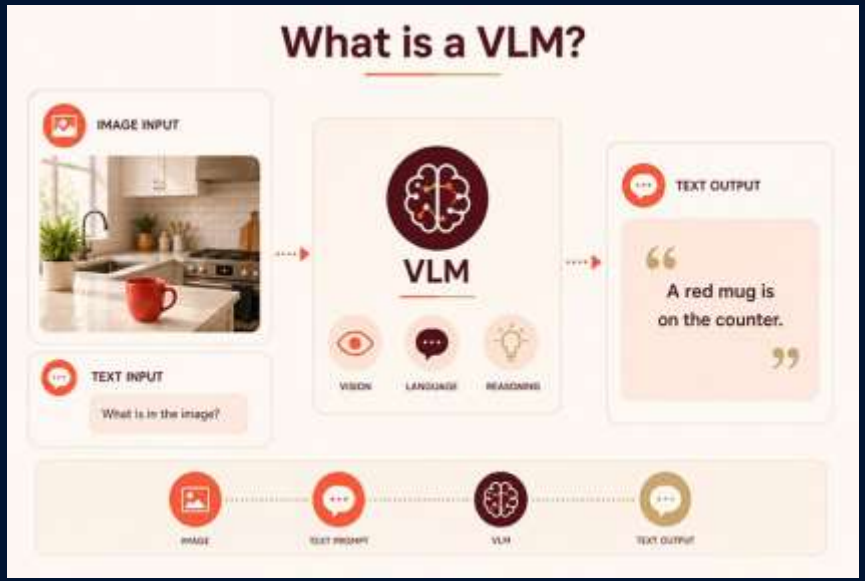
Summer 2026 _ University of Qom



Introduction

- Computers can process images.
- Computers can process natural language.
- But images and language are fundamentally different modalities.
- VLMs aim to connect these modalities.





Computer Vision

How can a machine understand an image?



- Computer vision is a subfield of artificial intelligence
- Process, analyze and interpret visual inputs such as images and videos
- An image is represented as numerical pixel values.
- Computer Vision extracts meaningful visual representations from these pixels.
- Traditional systems were usually designed for specific tasks.
- Classification / Detection / OCR



Natural Language Processing

How can a machine understand language?

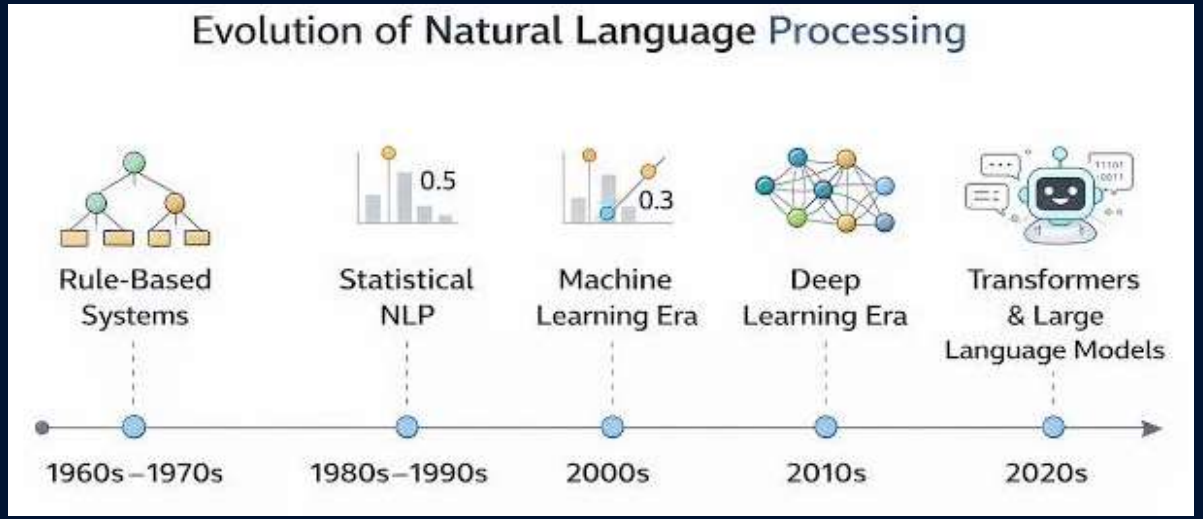


Natural language is sequential and symbolic.

NLP enable computers to understand and communicate with human language

NLP converts text into numerical representations.

These representations allow models to learn linguistic patterns.

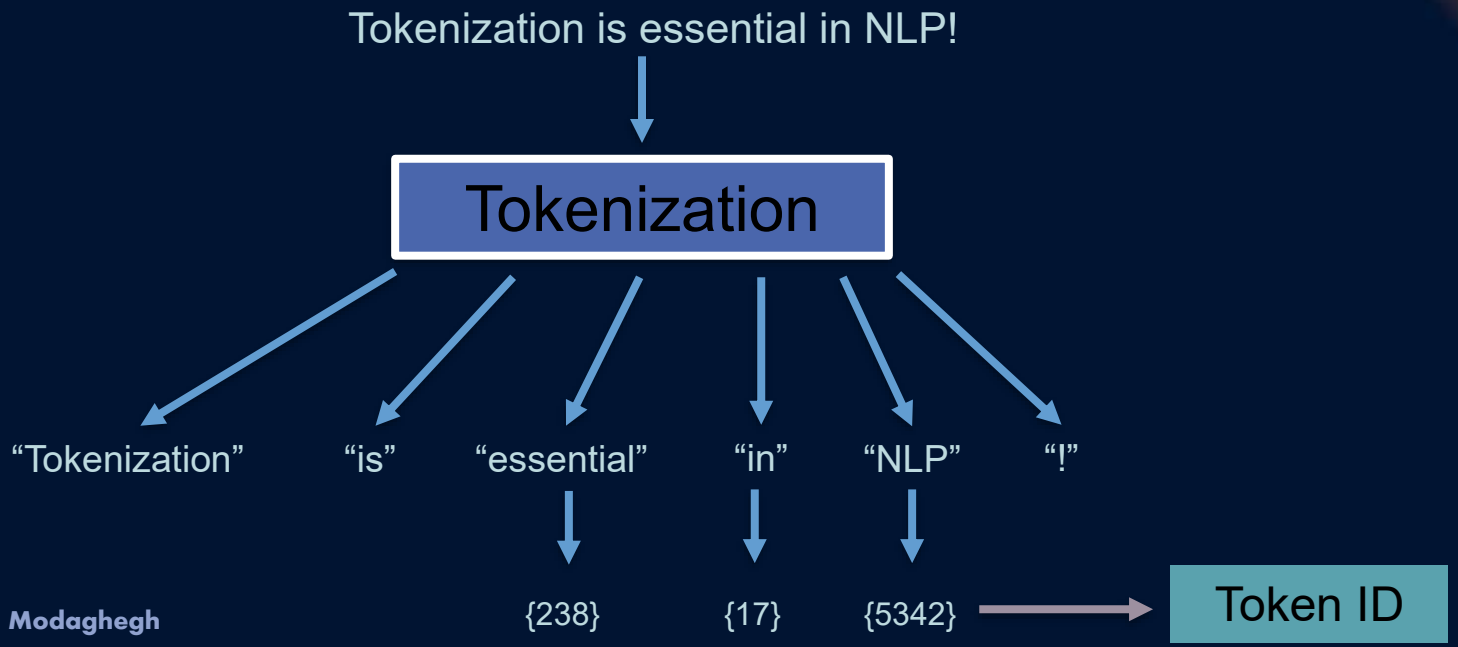


Tokenization

Neural networks cannot directly process raw text.

Tokenization converts text into discrete units called tokens.

Tokens are then mapped to numerical representations.(Token ID)



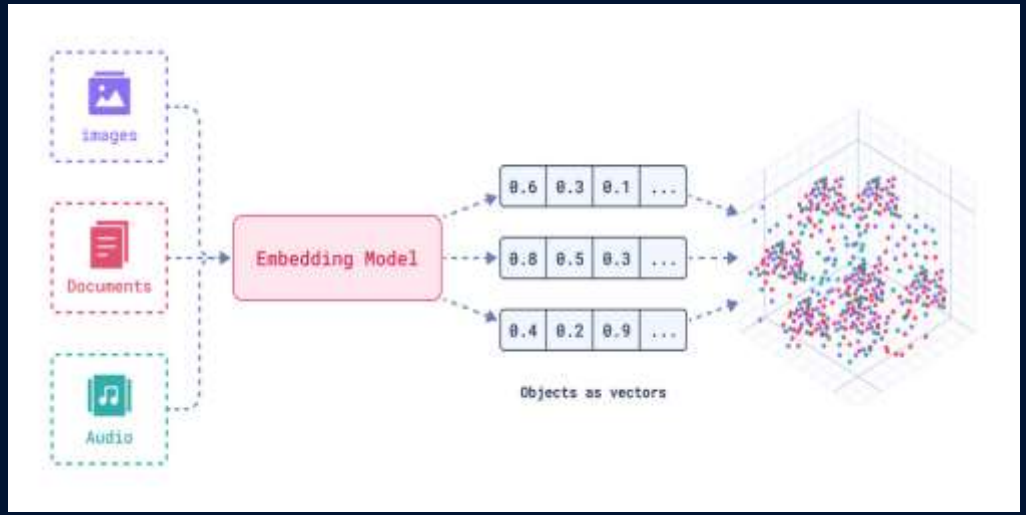
Embeddings

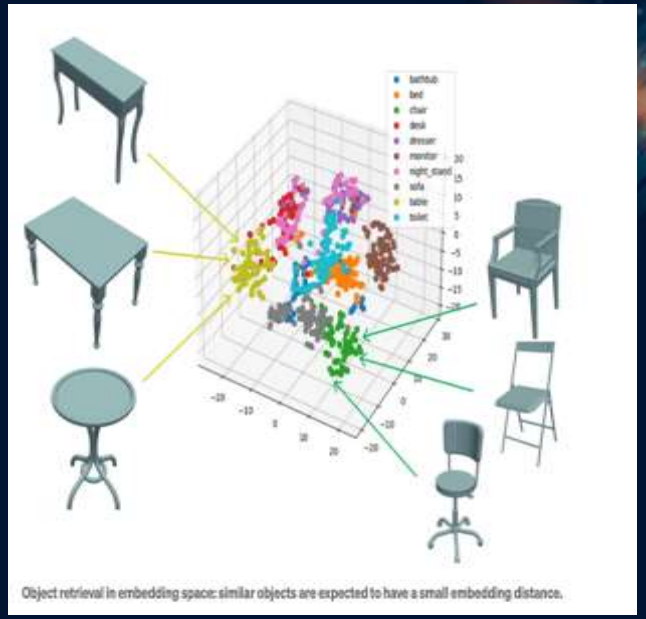
Tokens → Vectors

An embedding represents a token as a vector.

Similar meanings can have similar representations.

Embeddings provide a numerical semantic space for language.





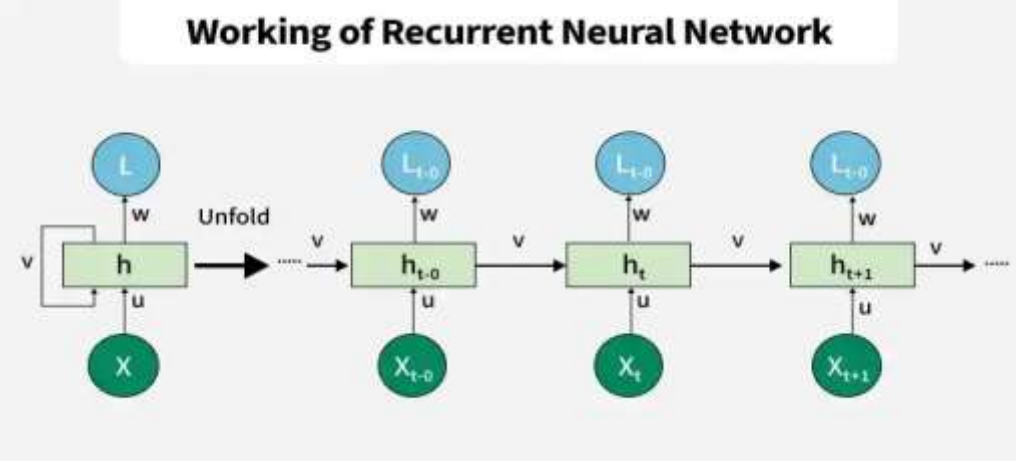
RNN

Recurrent Neural Network

RNNs are designed to process sequential data.

Key idea:

The previous hidden state is passed to the next time step, allowing the model to maintain information from the past.



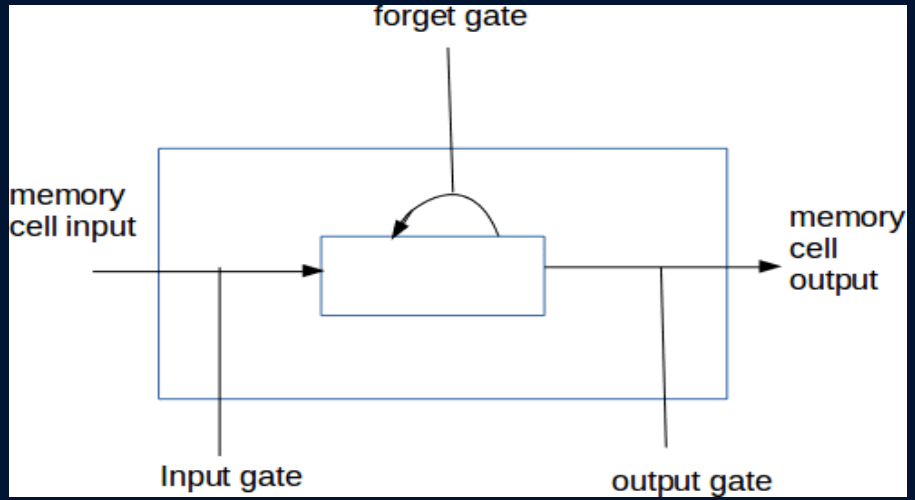
- **Sequential processing**
- **Hidden state as memory**
- **Shared parameters across time steps**

LSTM

Long Short-Term Memory

LSTM is a specialized RNN designed to learn long-term dependencies in sequential data. Instead of relying only on a hidden state, LSTM introduces:

- Cell State
- Forget Gate
- Input Gate
- Output Gate

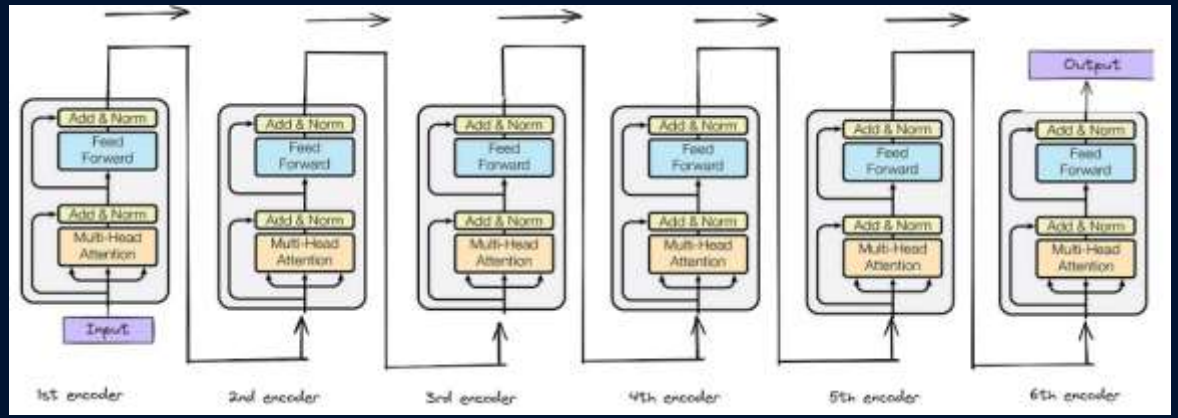


Main idea: Learn what to remember, what to forget, and what to output

Limitations of Sequential Architectures

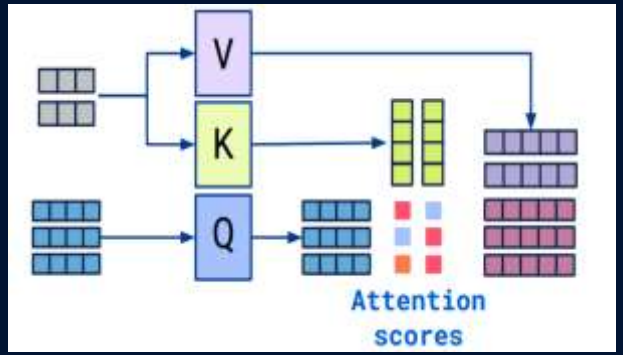
- Sequential computation limits parallelization.
- Long-range dependencies are difficult to preserve.
- Processing long sequences becomes inefficient.

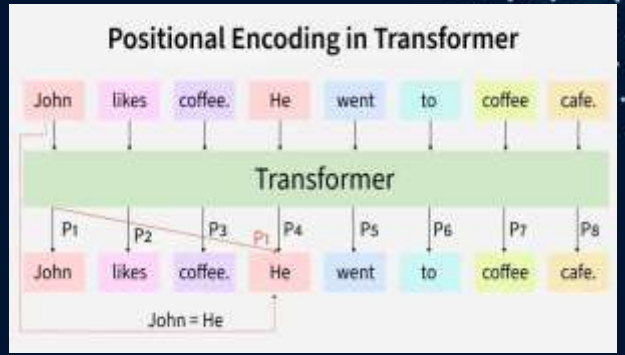
Can we process relationships between all tokens more directly?



Transformer

Transformer





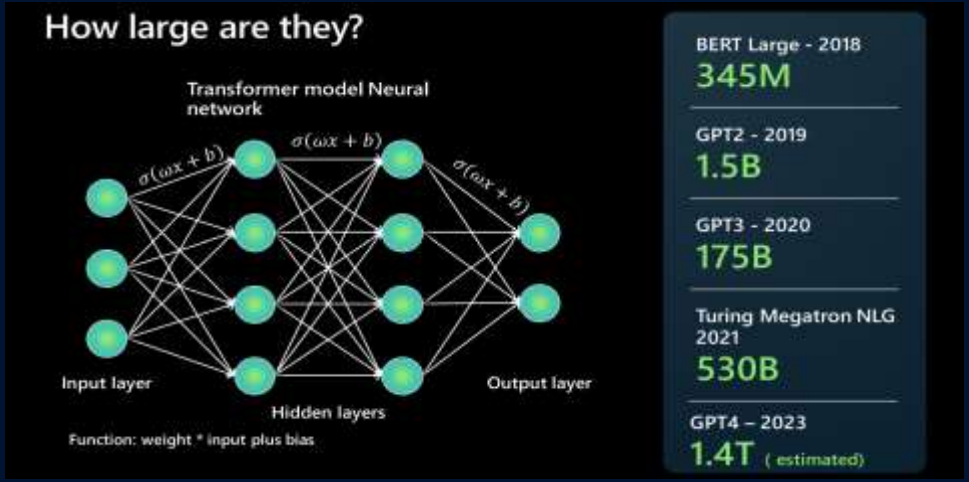
Input Tokens
↓
Embedding
↓
Positional Information
↓
Self-Attention
↓
Feed Forward Network
↓
Output Representations

- It allows tokens to interact with each other.
- Transformer removes the need for recurrent processing.
- It enables highly parallel computation.
- It became the foundation of modern LLMs.

LLM

Large Language Models

- LLMs are large Transformer-based language models.
- They are pre-trained on massive amounts of text.
- They learn linguistic patterns, knowledge, and reasoning capabilities.
- They can generate text autoregressively.



Why Can't We Directly Feed an Image to an LLM?

IMAGE		LLM
[Pixels]		[Tokens]
[Pixels]	≠	[Tokens]
[Pixels]		[Tokens]

Different modalities → Different representations

CLIP

Learning Transferable Visual Models From Natural Language Supervision

- **Authors:** Alec Radford et al.
- **Conference:** ICML 2021
- **Year:** 2021

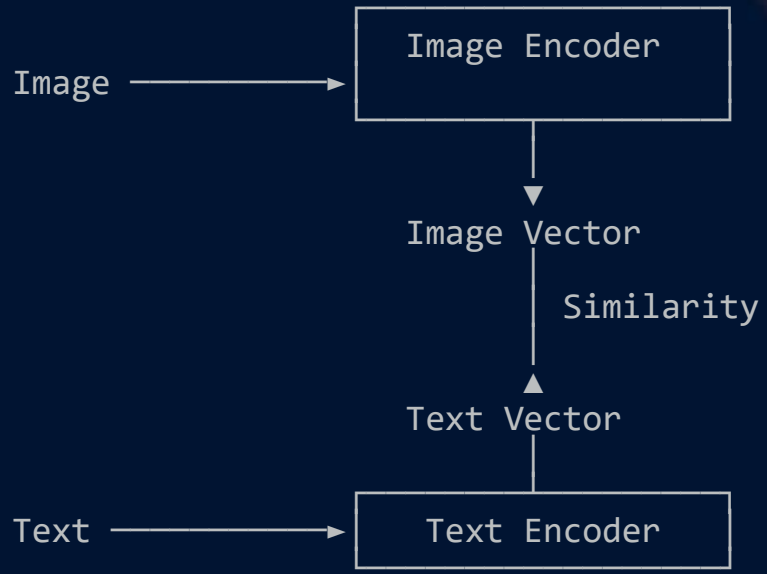
Problem:
Fixed visual categories and limited supervision

Key Idea:
Align images and natural language

Architecture:
Image Encoder + Text Encoder

Contribution:
Zero-Shot Transfer

Limitation:
Not a conversational image-to-text model



Flamingo

a Visual Language Model for Few-Shot Learning

- **Authors:** Jean-Baptiste Alayrac et al.
- **Conference:** NeurIPS 2022
- **Year:** 2022

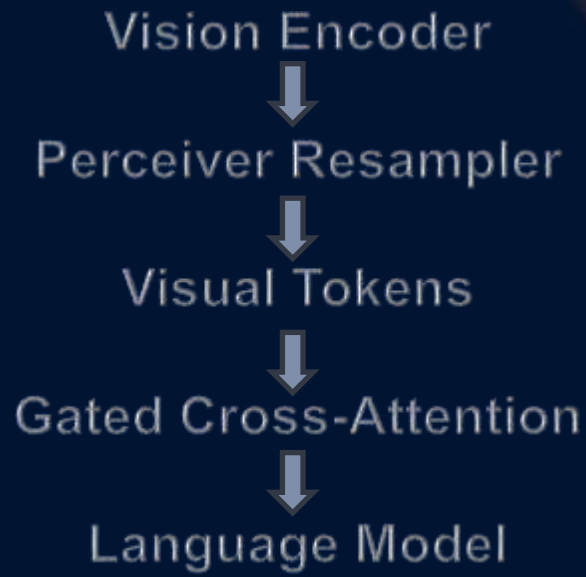
Problem
How to connect pretrained vision and language models?

Key Idea
Build a bridge between them

Architecture
Perceiver Resampler + Gated Cross-Attention

Contribution
Few-Shot Multimodal Learning

Limitation
Very large and computationally expensive



BLIP-2

Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models

- **Authors:** Junnan Li et al.
- **Conference:** ICML 2023
- **Year:** 2023

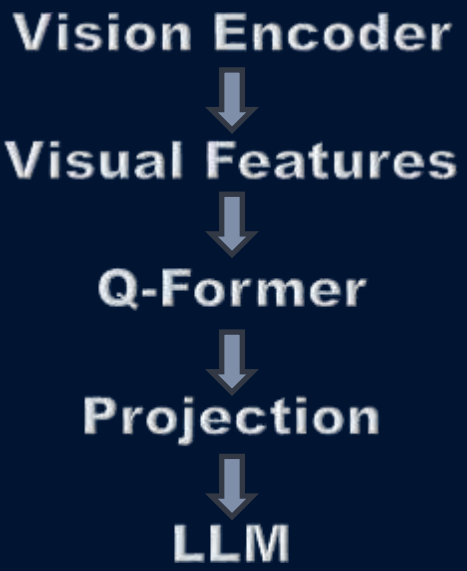
Problem
Expensive end-to-end training of large vision-language models

Key Idea
Use frozen pretrained models and learn a lightweight bridge

Architecture
Vision Encoder + Q-Former

Contribution
Efficient Vision-Language Learning

Limitation
Q-Former can limit fine-grained visual information



LLaVA

Visual Instruction Tuning

- **Authors:** Haotian Liu et al.
- **Conference:** NeurIPS 2023
- **Year:** 2023

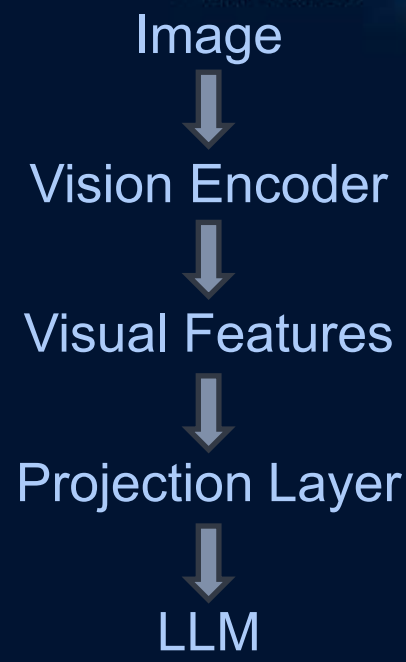
Problem
How to make a vision-language model follow user instructions?

Key Idea
Connect a pretrained vision encoder to an LLM using a projection layer

Architecture
Vision Encoder + Projection + LLM

Contribution
Visual Instruction Following

Limitation
Limited visual understanding compared with specialized models



Qwen-VL

A Versatile Vision-Language Model for Understanding, Localization, Text Reading, and Beyond

- **Authors:** Jinze Bai et al.
- **Base Model:** Qwen-LM
- **Year:** 2023

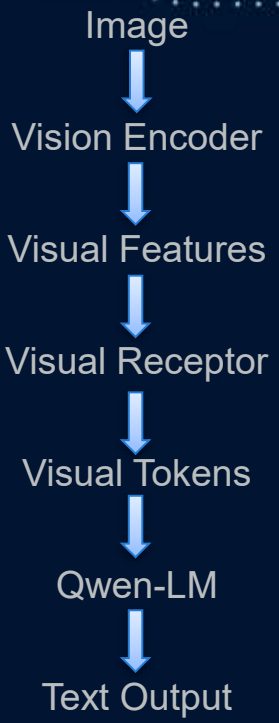
Problem
How to achieve fine-grained visual understanding?

Key Idea
Connect visual features to Qwen-LM and train for diverse visual tasks

Architecture
Vision Encoder + Visual Receptor + Qwen-LM

Contribution
OCR, Grounding, Localization and Visual Chat

Limitation
Larger and more complex than lightweight VLMs



InternVL

Scaling up Vision Foundation Models and Aligning for Generic Visual-Linguistic Tasks

- **Authors:** Zhe Chen et al.
- **Conference:** CVPR 2024
- **Year:** 2024

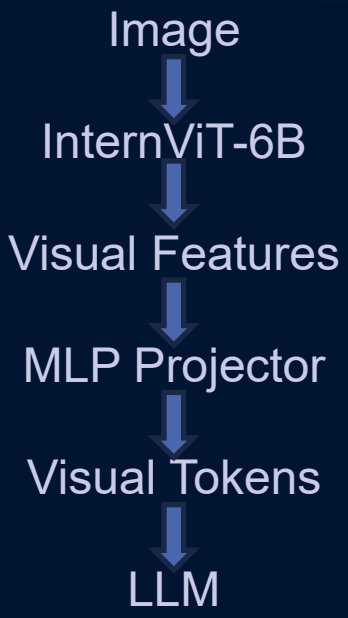
Problem
How can we improve visual understanding by scaling the vision component?

Key Idea
Scale up the Vision Encoder and align it with an LLM

Architecture
InternViT + MLP Projector + LLM

Contribution
Strong Visual-Linguistic Understanding

Limitation
High computational and hardware requirements



Model Comparison

Model	Main Question	Key Mechanism	Main Capability
CLIP	How can vision and language be aligned?	Dual Encoder + Contrastive Learning	Zero-Shot
Flamingo	How can pretrained vision and language models interact?	Resampler + Cross-Attention	Few-Shot Multimodal Learning
BLIP-2	How can this connection be made more efficiently?	Q-Former	Efficient VLM
LLaVA	How can a VLM follow user instructions?	Vision Projection + LLM	Visual Chat
Qwen-VL	How can a VLM understand fine-grained visual information?	Visual Receptor + LLM	OCR / Grounding
InternVL	How far can visual capability be scaled?	Large Vision Encoder + Alignment	Strong Visual Understanding



Thank You

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